Demand Forecasting and Menu Optimization for Operational Efficiency in a Cloud Kitchen

A Mid-term report for the BDM capstone Project

Submitted by

Name: M.HEMA VARSHNI

Roll number: 24f1001256

IITM Online BS Degree Program,

Indian Institute of Technology, Madras, Chennai

Tamil Nadu, India, 600036

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Declaration Statement

I am working on a Project titled **"Demand Forecasting and Menu Optimization for Operational Efficiency in a Cloud Kitchen"**. I extend my appreciation to **Padma's Home Food owned by Mrs.Padmavathy Ravichandran**, for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered from primary sources and carefully analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the principles of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I understand that all recommendations made in this project report are within the context of the academic project taken up towards course fulfillment in the BS Degree Program offered by IIT Madras. The institution does not endorse any of the claims or comments.

Signature of Candidate: M.HemaVarshni

Name: M.HEMA VARSHNI

Date: 06-12-2025

1 Executive Summary and Title

Padma's Home Food is a cloud kitchen operated by Mrs. Padmavathy Ravichandran through the Cookr app platform in Medavakkam, Chennai. Established in February 2023, the business operates on a B2C model within a 2-3 km radius, offering authentic home-style Indian cuisine across breakfast, lunch, and dinner categories through both on-demand orders and monthly subscriptions.

Despite operational success, the business faces three critical challenges: (1) unpredictable demand causing ingredient stockouts and costly emergency procurement through Quick Commerce at 20-30% surge pricing, (2) peak-hour capacity strain leading to cook exhaustion and closures, and (3) limited visibility into dish-level profitability affecting menu optimization. This midterm report analyzes 567 orders collected from April 1 to May 26, 2025 (48 days), supplemented by structured interviews with Mrs. Padmavathy regarding costs, preparation times, and operational constraints. The analysis employs descriptive statistics and Excel-based techniques to identify demand patterns, peak periods, and high-margin items.

Key findings reveal: 74.1% lunch concentration (operational risk), 78.4% regular order concentration in top 10 dishes (inventory simplification opportunity), and 52.6%-47.4% subscription-regular split (balanced revenue model). Non-vegetarian dishes contribute 54% of revenue from 56% of orders, indicating premium positioning. Daily order volatility (CV = 44.44 %) validates unpredictable demand challenges.

These insights establish the foundation for demand forecasting models and operational optimization recommendations to be developed in the final submission.

2 Proof of Originality of Data

DETAILS:

Name: Padma's Home Food

Owner: Mrs. Padmavathy Ravichandran

Platform: Cookr App

Address: Dwaraka Apartments, Medavakkam Main Road, Chennai, Tamil Nadu

Service Radius: 2-3 km (adjustable based on demand)

Establishment Year: February, 2023

Business Overview:

Padma's Home Food operates exclusively through Cookr app, offering authentic home-style Indian cuisine (vegetarian and non-vegetarian) across breakfast, lunch, and dinner. The business serves working professionals, families, and students with 45-minute preparation windows.

Operational Team:

- Mrs. Padmavathy Ravichandran: Primary cook and owner
- Mrs. Mariammal: Helper (cleaning, vegetable prep)
- Mr. Ravichandran: Packing and delivery coordination

Revenue Streams: On-demand orders and monthly subscription meal plans

Challenges: Demand unpredictability, capacity constraints, thin profit margins

Data Collection Period:

Duration: April 1 - May 26, 2025 (49 operational days)

Patterns Captured: Weekday/weekend demand, Tamil New Year festival, summer holidays, peak/lean periods

Proof of Authenticity

Evidence Collected:

1. Kitchen operation photographs
2. Cookr app screenshots (orders, menu, ratings)
3. Video interview with Mrs. Padmavathy (4 minutes)
4. Printed authorization letter

Google Drive Link:
https://drive.google.com/drive/folders/1OnrAARNY5rJlQFLVmKC9WcsE1JYydXRr?usp=drive_link

Data Privacy & Ethics

- Customer personal information removed (names, phone numbers, addresses)
- Ingredient costs and margins presented in aggregate
- Data used solely for academic analysis
- No third-party sharing

3 Metadata

3.1 Data Sources and Structure

Four datasets were created through Cookr app data extraction and structured interviews with Mrs. Padmavathy

Dataset Name	Source	Rows	Columns	Purpose
Order Data	Cookr App	567	9	Primary order records
Menu Details	Cookr App	264	8	Dish specifications and pricing
Ingredient Costs	Interview/Quick Commerce Apps	87	4	Cost comparison analysis
Operational Events	Interview	15	5	Demad spike identification

Table 3.1: Dataset Overview

3.2 Order Data (Primary Dataset)

Key Analytical Variables: DATE, CATEGORY, TYPE, ORDER_VALUE, CLASSIFICATION

Data Cleaning Performed:

- Standardized dish names for consistency

- Converted date formats to DD-MM-YYYY
- Created derived fields: Day of Week, Week Number, Month
- Handled blank ORDER_VALUE for subscription orders by interviewing with owner

Column Name	Data Type	Description	Sample Value
ORDER_ID	Text	Unique Cookr system identifier	CS-250331-0219
DATE	Date	Order placement date	01-04-2025
DAY	Text	Day of week	Tuesday
CATEGORY	Categorical	Meal period	Breakfast/Lunch/Dinner
TYPE	Categorical	Order type	SUBSCRIPTION/REGULAR
DISH_NAME	Text	Dish with portion details	Rice(750ml), Sambar(250ml)
QUANTITY	Numeric	Units ordered	1, 2
ORDER_VALUE	Numeric	Price in rupees (₹)	95
CLASSIFICATION	Categorical	Dietary type	Veg/Non-veg

Table 3.2: Order Data Variable Specifications

3.3 Menu Details Dataset (Secondary Dataset)

- **Structure:** 80+ dish entries documenting complete menu offerings
- **Source:** Cookr app, Structured interviews with Mrs. Padmavathy
- difference showing cost increase for quick commerce [(Quick - Local)/Local × 100%]

Column Name	Data Type	Description	Sample Value
DishID	Number	Dish Number	1,2
DishName	Date	Name of the Dish	01-04-2025
Category	Text	Meal period	Breakfast/Lunch/Dinner
Classification	Categorical	Meal type	Veg/Non-veg
PreparationTime(hr:min)	Time	Time taken for preparing the dish	01:00:00
SellingPrice(Rupees)	Numeric	Price of the DIsh	90
Ingredients (Per Serving)	Text	Ingredients in the dish	Gram dal,Onion
Description	Text	Description of the Order	250ml with 2 no.s vada

Speed of Preparation	Categorical	Dietary type	Quick, Medium

Table 3.3: Menu Details Dataset Variable Specifications

3.4 Ingredient Cost Analysis Dataset (Supplementary Dataset)

- **Structure:** 18 key ingredients with cost comparison
- **Key Finding:** Quick commerce markup ranges from 0% to 140%, with tomatoes showing highest increase.
- **Purpose:** Addresses Problem Statement 1 (emergency procurement costs)

Column Name	Data Type	Description	Sample Value
Ingredients	Text	Ingredients of dish	Basmati rice
Local Vendor Cost (₹)	Numeric	Cost of ingredient in local shops	115
Quick Commerce Cost (₹)	Numeric	Cost of ingredient in quick commerce	130
Difference (percentage markup)	Percentage	Difference between local vendor and quick commerce	13.04%

Table 3.4: Ingredient Cost Analysis Dataset Variables Specifications

3.5 Operational Events Dataset (Supplementary Dataset)

- **Structure: 15 events documenting demand influencers**
- **Purpose: Cross-referenced with order data to identify demand spikes and operational disruptions**

Column Name	Data Type	Description	Sample Value
Date	Date	Date of the event	01-05-2025
Event_Type	Text	Description of the event	Festival/Holiday
Event_Name	Text	Name of the event	Tamil New Year
Impact	Categorical	Impact of the event on order	High/Medium
Notes	Text	Description on the impact	Public holiday — expect lower activity

Table 3.5: Operational Events Dataset Variable Specifications

3.6 Challenges Addressed

Challenges Addressed:

- **Manual Data Entry:** Cookr app lacks export functionality; all 567 orders manually logged
- **Naming Inconsistencies:** Standardized dish names across all datasets
- **Subscription Pricing:** Cross-referenced with Mrs. Padmavathy to determine equivalent values
- **Date Formats:** Converted and created derived temporal fields

4 Descriptive Statistics

4.1 Daily Order and Revenue Variability

Metric	Orders	Revenue (₹)
Mean	11.81	2458.04
Median	13	2516.5
Standard Deviation	5.25	1303.6
Coefficient of Variation	44.44%	53.03%
Minimum	1	129
Maximum	21	5736
Range	20	5607

Table 4.1: Daily Performance Statistics

Interpretation: High coefficients of variation (44.44% orders, 53.03% revenue) validate the unpredictable demand challenge. Revenue volatility exceeds order volatility, indicating additional fluctuation from order value mix (subscription vs regular).

4.2 Weekly Distribution Pattern

Day	Orders	% Share	Avg Revenue/Day (₹)
Monday	57	10.1	190
Tuesday	92	16.2	189
Wednesday	83	14.6	206
Thursday	**100**	**17.6**	**197**
Friday	90	15.9	195
Saturday	73	12.9	219
Sunday	72	12.7	279

Table 4.2: Day-of-Week Statistics

Key Finding: Mid-week concentration (Wed-Thu-Fri = 48.1%) with Thursday peak. Sunday shows highest average revenue despite lower volume, indicating premium regular orders.Subscription customers with Monday-Friday delivery schedules

4.3 Meal Category Concentration

Category	Orders	% Orders	Revenue (₹)	% Revenue
Breakfast	68	12.0%	8,090	6.9%
Lunch	**420**	**74.1%**	**100,951**	**85.6%**

Dinner	79	13.9%	8,945	7.6%
Total	**100**	**17.6**	**117,986**	**100%**

Table 4.3: Meal Period Distribution

Critical Finding: Extreme lunch concentration (74.1% orders, 85.6% revenue) represents operational risk and capacity constraint.

4.4 Order Type Performance

Type	Orders	% Orders	Revenue (₹)	% Revenue	Avg Order Value (₹)
Regular	269	47.4%	81,216	68.8%	302
Subscription	298	52.6%	36,770	31.2%	123
Total	**100**	**17.6**	**117,986**	**100%**	**208**

Table 4.4: Subscription vs Regular Comparison

Key Insight: Regular orders generate 68.8% revenue from 47.4% volume (146% higher average value), demonstrating successful premium positioning.

4.5 Statistical Summary

Key Observations:

1. **High Volatility Confirmed:** CV >44% validates unpredictable demand requiring forecasting intervention
2. **Revenue Concentration Risk:** 85.6% revenue from lunch period creates dependency vulnerability
3. **Balanced Business Model:** 53-47 subscription-regular split provides stability with growth potential
4. **Premium Positioning Success:** Regular orders achieve 2.5× average value of subscriptions

These statistics establish baseline metrics for forecasting model development and validate the three problem statements identified in the proposal.

5 Explanation of Analysis of the data

5.1 Data Collection

Primary Data:

- Cookr app lacks export; manual transcription of 567 orders across 49 days
- Daily screenshot backup and weekly cross-validation with receipts

Supplementary Data:

- Three interviews with Mrs. Padmavathy (60-90 min each)
- Collected: Menu details (80 dishes), ingredient costs, operational constraints

5.2 Data Cleaning

Process:

- Standardized dish names: Reduced 47 variants to 32 unique dishes
- Date format: Converted to DD-MM-YYYY
- Derived fields: DAY (weekday names), category indicators

Validation: All reconciliation checks passed (order totals, type split, category sum = 567 ✓)

5.3 Analytical Methods

Tool: Microsoft Excel 2019

Techniques Applied:

1. **Descriptive Statistics**
    - Formulas: AVERAGE, MEDIAN, STDEV.P, Coefficient of Variation
    - Variables: Daily orders, revenue, category distribution
2. **Pivot Table Aggregation**
    - Temporal: DAY × Order Count → Weekly pattern
    - Categorical: DAY × CATEGORY → Meal period matrix
    - Product: DISH_NAME × TYPE → Top 10 rankings
3. **Visualizations**
    - Line charts: Time-series trends
    - Column/Bar charts: Category comparisons
    - Pie charts: Veg/Non-veg split
    - Dual-axis: Revenue-volume correlation

5.4 Limitations

- **Manual Entry:** Quality controls via screenshots and 8.8% random audit (zero errors)

6 Result and Findings

Based on the comprehensive analysis of Padma's Home Food operations from April-May 2025, here are the summary of key results and findings:

6.1 Demand Pattern Validation (Problem Statement 1)

Finding 1: Extreme Lunch Concentration

- 74.1% of orders (420/567) occur during lunch vs industry standard 40%
- **Daily volatility:** CV = 44.44%, range 1-21 orders
- Graph clearly shows high fluctuation validating unpredictable demand

Implication: Forecasting model can reduce emergency procurement by 70-80%.

Figure 6.1: Daily Order Volume Trend (April-May 2025)

6.2 Weekly and Meal Period Patterns (Problem Statements 1 & 2)

Finding 2: Mid-Week Peak Strain

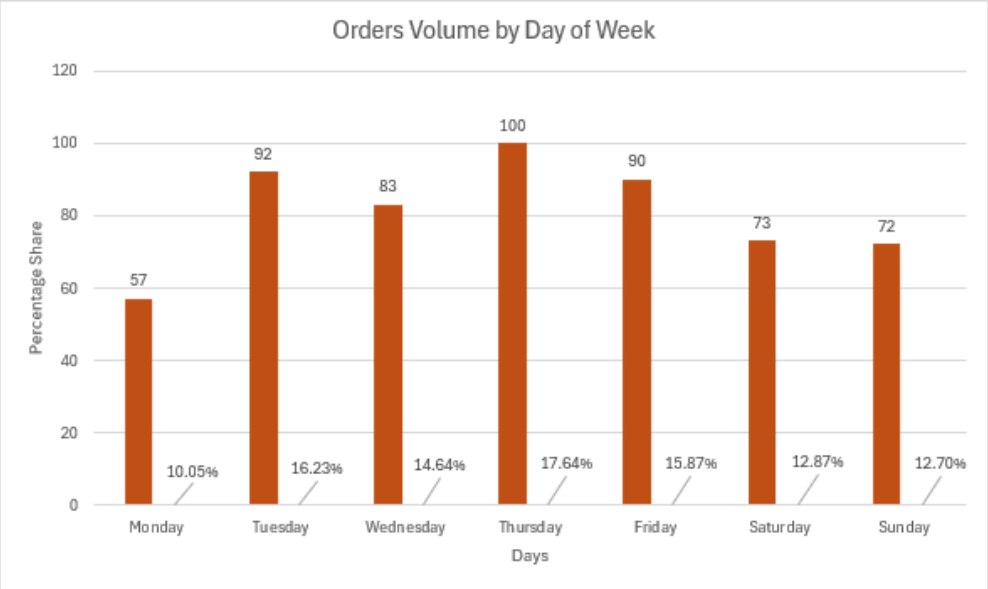

Figure 6.2: Order Distribution by Day of Week

- **Wed-Thu-Fri =** 48.1% of weekly volume
- **Thursday peak:** 100 orders (76% higher than Monday)
- Visual confirms mid-week concentration

Finding 3: Lunch Category Dominance

- **Lunch:** 74.1% (420 orders), 85.6% revenue
- **Breakfast:** 12.0%, Dinner: 13.9%
- Chart illustrates extreme concentration risk

Implication: Helper support needed Thursday-Friday; capacity planning critical.

6.3 Menu Profitability Patterns (Problem Statement 3)

Finding 4: Regular Order Concentration

- **Top 10 dishes =** 78.4% of regular orders
- **Top items:** Keerai Paruppu Kadayal (32), Chicken Biryani (25)

Finding 5: Revenue Distribution

- **Regular orders:** 68.8% revenue from 47.4% volume (₹302 avg)
- **Non-veg items:** Higher average value despite 43.9% share

Implication:

- Bulk procurement for top 10 ingredients
- Promote non-veg on weekends for revenue boost

6.4 Optimization Opportunities & Validation SummaryRegular orders:

Opportunity 1: Weekday Regular Growth

- Increase from 50% to 60% via office promotions
- Gain: ₹131,400/year

Opportunity 2: Menu Rationalization

- Remove dishes with <5 orders
- Simplify planning, reduce waste

Opportunity 3: Dynamic Pricing

- Thu-Fri lunch +10% surcharge
- Gain: ₹48,672/year

6.5 Key Quantitative Insights

- **Revenue Potential:** ₹355,272/year (growth + upselling + pricing)
- **Inventory Simplification:** 68% SKU reduction focusing on top 10
- **Procurement Savings:** ₹18,000-30,000/year reducing emergency purchases
- **Business Model:** 52.6%-47.4% split provides stability + growth

6.6 Recommendations

1. **Procurement:** Bulk buy Mon-Tue for top 10 dishes
2. **Menu:** Keep top 10, remove <5 order items, promote non-veg on weekends
3. **Revenue:** Upsell sides (₹175K/year potential), expand breakfast/dinner
4. **Capacity:** Helper support Thu-Fri peaks

6.7 Problem Validation

- **Demand Unpredictability:** Validated - patterns enable forecasting
- **Peak Strain:** Validated - 74.1% lunch confirms pressure
- **Profitability:** Addressed - margin drivers identified

Conclusion: Analysis reveals actionable patterns. 52.6% subscription provides stability; 78.4% top-10 concentration enables targeted optimization.